

PRITAM SARKAR

Rampura, Dhaka, Bangladesh

pritamsarkar.prs@gmail.com | +880 1937030596

linkedin.com/in/pritam-sarkar-5aba07212 | GitHub | Google Scholar | ORCID:
0009-0006-1297-4760

PROFESSIONAL SUMMARY

Fresh Computer Science undergraduate from BRAC University seeking entry-level and internship opportunities across multiple sectors. Strong foundation in machine learning, data analysis, and software development, with hands-on experience in geospatial, time-series, and multimodal data processing. Technical skills include Python, PyTorch, TensorFlow, SQL, data analysis, and end-to-end MLOps pipelines, with practical experience in cloud deployment and full-stack development. Published researcher (IEEE, Springer) with experience in model development, evaluation, and technical documentation. Demonstrates strong communication, teamwork, problem-solving, and organizational skills. Adaptable, detail-oriented, and committed to delivering high-quality work in both technical and general professional roles. Open to diverse job functions and industries, with a focus on continuous learning and professional growth.

WORK EXPERIENCE

Information Technology Intern

Aug 2026 – Present

Modhumoti Bank PLC.

Internship · Gulshan, Dhaka, Bangladesh (On-site)

- Provide technical support and troubleshoot hardware, software, and system-related issues.
- Assist with IT infrastructure, network monitoring, and system maintenance and configuration.
- Support documentation and routine IT security procedures.
- Collaborate with the IT team to ensure smooth daily operations.

Undergraduate Researcher (Thesis & Independent Research)

Jan 2025 – Jan 2026

BRAC University, Dhaka, Bangladesh

- Led development of a multi-stage deep learning framework for automated detection and localization of shrimp disease as part of undergraduate thesis research.
- Co-authored three additional peer-reviewed papers on satellite-driven evapotranspiration/drought analysis, soil moisture recharge prediction, and wind energy resource mapping, published/presented via IEEE and Springer.
- Presented research findings at three international conferences (IEEE COMPAS 2025; BIM 2025; RAAICON 2026).

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science

Jan 2022 – Apr 2026

BRAC University, Dhaka, Bangladesh

GPA: 3.30 / 4.00

Higher Secondary Certificate (HSC) – Science

Jul 2018 – Dec 2020

Dhaka City College, Dhaka, Bangladesh

GPA: 5.00 / 5.00

Secondary School Certificate (SSC) – Science

Jan 2013 – Apr 2018

Motijheel Government Boys' High School, Dhaka, Bangladesh

GPA: 5.00 / 5.00

RESEARCH & PUBLICATIONS

A Multi-Stage Deep Learning Framework for Automated Detection and Localization of Shrimp Disease

P. Sarkar, M. T. Hasan, A. Saha, M. M. Siraj, M. M. Abdullah

Undergraduate Thesis, BRAC University, January 2026

Repository: <http://hdl.handle.net/10361/28080> | Full Paper

Enhanced Unsupervised Machine Learning Analysis on Evapotranspiration and Drought Based on Satellite-Driven Data

P. Sarkar, T. Paul

3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025), Dhaka International University, Bangladesh, September 2025

Lecture Notes in Networks and Systems, vol. 1800, Cham: Springer, 2026, pp. 507–520

A Hybrid Attention-Based ST-CNN and Penman-Monteith Framework for High-Resolution Soil Moisture Recharge Prediction Using Satellite-Driven Data

T. Paul, P. Sarkar

IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025), Islamic University, Kushtia, Bangladesh, October 2025

<https://ieeexplore.ieee.org/abstract/document/11381752>

Nationwide Wind Energy Resource Mapping in Bangladesh Using Self-Organizing Maps (SOM): A Machine Learning-Based Site Suitability Assessment

T. Paul, P. Sarkar

International Conference on Big Data, IoT and Machine Learning, Springer, 2025 (pp. 415–429)

https://link.springer.com/chapter/10.1007/978-3-032-15346-3_29

CONFERENCES & PRESENTATIONS

Presenter

- “A Multi-Stage Deep Learning Framework for Automated Detection and Localization of Shrimp Disease,” IEEE 5th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026), Jashore University of Science and Technology, Jashore, Bangladesh, September 2026. [Certificate]
- “A Hybrid Attention-Based ST-CNN and Penman-Monteith Framework for High-Resolution Soil Moisture Recharge Prediction Using Satellite-Driven Data,” IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025), Islamic University, Kushtia, Bangladesh, October 2025. [Certificate]
- “Nationwide Wind Energy Resource Mapping in Bangladesh Using Self-Organizing Maps (SOM): A Machine Learning-Based Site Suitability Assessment,” 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025), Dhaka International University, Bangladesh, September 2025. [Certificate]
- “Enhanced Unsupervised Machine Learning Analysis on Evapotranspiration and Drought Based on Satellite-Driven Data,” 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025), Dhaka International University, Bangladesh, September 2025. [Certificate]

PROJECTS

DeepResearch AI

GitHub Repository

Multi-Agent Systems · LangGraph · RAG · ChromaDB · LLM Tool Use · Streamlit

Built a full-stack AI research assistant using a 4-agent pipeline (Planner → Researcher → Critic → Synthesizer) orchestrated with LangGraph. The system accepts any research topic, retrieves context from uploaded PDF documents via RAG (ChromaDB and Sentence Transformers) as well as the live web (Wikipedia and DuckDuckGo), self-critiques its findings, and generates a fully written, cited research report with one-click PDF download – powered entirely by free tools including Groq’s Llama 3.1 and Streamlit. Demonstrates proficiency in RAG pipelines, multi-agent state machines, vector databases, LLM tool use, and production UI deployment on Streamlit Cloud, with full local usability.

AutoRecruiter – Local AI Recruitment Automation Agent

GitHub Repository

OCR · Local LLMs (Ollama) · ChromaDB · Semantic Search · Streamlit

Developed a fully local AI-powered recruitment automation platform that parses resumes (PDF, DOCX, and scanned images), extracts structured candidate information using OCR and local LLMs, and ranks applicants through semantic similarity, skills matching, and AI-based evaluation. Generates customized interview questions per candidate and stores profiles in a searchable ChromaDB vector database. Built with Python, Streamlit, Ollama, Tesseract OCR, and ChromaDB, ensuring complete data privacy by eliminating reliance on cloud APIs.

Agri-Scan AI

GitHub Repository | Live Demo

Deep Learning · Computer Vision · EfficientNet-B0 · ONNX · Streamlit

Intelligent agricultural diagnostic web application that detects 22 crop disease phenotypes across Cashew, Cassava, Maize, and Tomato. Utilizes a fine-tuned EfficientNet-B0 architecture optimized for edge deployment via ONNX. Pipeline incorporates CLAHE-based image normalization and aggressive geometric

augmentation for robustness in diverse field conditions. Demonstrates end-to-end MLOps proficiency from automated data sanitization and model optimization to a live, user-centric web interface hosted on Streamlit Cloud.

Air Touch Control System

GitHub Repository

Computer Vision · MediaPipe · OpenCV · Python · HCI

Real-time touchless desktop interaction system using hand gesture recognition via OpenCV and MediaPipe's 21-point hand tracking. Key engineering contributions include: an Exponential Moving Average (EMA) filter ($\alpha = 0.25$) to eliminate camera jitter with sub-15ms latency; a state-machine-based drag-and-drop engine to prevent click-looping; and a virtual joystick scroll/zoom mechanism using a mid-air coordinate anchor to mitigate hand fatigue. Supports precision cursor control, left/right click, continuous scroll, and pinch-to-zoom. Packaged as a standalone Windows executable via PyInstaller for zero-dependency deployment.

Audio-Lyrics Clustering

GitHub Repository | Project Report

Deep Learning · Multimodal Fusion · Python · PyTorch / TensorFlow

Implemented a deep learning pipeline to cluster music tracks by fusing Audio Spectrograms with Lyrics embeddings. Applied multimodal feature extraction – mel-spectrograms for audio and transformer-based encoders for lyric text – followed by joint representation learning and unsupervised clustering. Demonstrates expertise in multimodal AI, signal processing, and NLP applied to real-world unstructured data.

TECHNICAL SKILLS

Programming Languages

Python (Advanced), JavaScript (ES6+), C++, SQL, Bash / Shell Scripting, HTML, CSS

AI & Deep Learning

CNNs (ResNet, EfficientNet, ConvNeXt-Tiny), RNNs, LSTMs, GRUs; Computer Vision (Image Classification, Object Detection, Semantic Segmentation, Grad-CAM); Spatiotemporal Forecasting; Explainable AI / XAI (SHAP, LIME, Integrated Gradients); Multi-Agent Systems (LangGraph), Retrieval-Augmented Generation (RAG); PyTorch, TensorFlow, Keras, Scikit-learn

Data Science & Statistical Analysis

Exploratory Data Analysis (EDA), Feature Engineering, Hypothesis Testing, ANOVA, Bayesian Inference, Regression (Linear / Logistic / Polynomial), SMOTE; NASA POWER weather data processing, Geospatial & Remote Sensing analysis; Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly

Data Analytics Software

Jupyter Notebook, Google Colab, Microsoft Excel (Advanced – PivotTables, VLOOKUP, data modelling), Tableau, Power BI

MLOps & Pipelines

ETL pipelines, Data Ingestion & Validation; Git, DVC, MLflow; Docker, Flask, FastAPI; Apache Airflow (Basic), Prefect, GitHub Actions; AWS S3, Heroku, Netlify; CI/CD for ML

Data Engineering

PostgreSQL, MySQL, MongoDB, ChromaDB, Pinecone (Vector Databases); Advanced SQL, Query Optimization, Data Warehousing concepts; RESTful API development

Software Engineering & Web Development

MERN Stack (MongoDB, Express.js, React.js, Node.js); OOP, MVC Architecture, Microservices; Agile / Scrum, Test-Driven Development (TDD); Git, GitHub, Bitbucket, Docker, Postman, PyCharm, VS Code

Networking & Cybersecurity

TCP/IP Suite, OSI Model, DNS, DHCP, Subnetting, Routing & Switching; Firewalls, VPNs, SSL/TLS, SSH, Encryption Standards; Cisco Packet Tracer, Wireshark, GNS3; AWS (Basic)

Operating Systems

Linux / Ubuntu (CLI, Bash Scripting, system administration), Windows 10/11

General Computer & Productivity Skills

Microsoft Office Suite (Word, Excel, PowerPoint), Google Workspace (Docs, Sheets, Slides, Drive, Co-

CERTIFICATIONS

- **IELTS Academic – Overall Band Score: 7.0 (C1)**
British Council · Candidate ID: A03330405
Listening: 8.0 · Reading: 7.0 · Writing: 6.5 · Speaking: 6.0
View Certificate
- **Associate AI Engineer for Data Scientists**
DataCamp · Jun 2026 · Credential ID: 8144b1e39d1cbb09b26c2328234401d640132c94
Verify Certificate
- **Supervised Machine Learning: Regression and Classification**
DeepLearning.AI & Stanford University · Aug 2025 · Credential ID: YAEEO7ZCOZOT
Verify Certificate
- **Connect and Protect: Networks and Network Security**
Google · Nov 2024 · Credential ID: PS20M704EGZI
Verify Certificate
- **Google AI Essentials**
Google · Nov 2024 · Credential ID: 0K6PNPZ35RX9
Verify Certificate
- **Databases and SQL for Data Science with Python (with Honors)**
IBM · Nov 2024 · Credential ID: UDJERO3UMJ71
Verify Certificate
- **Play It Safe: Manage Security Risks**
Google · May 2024 · Credential ID: 92DQW62K5DJX
Verify Certificate
- **Foundations of Cybersecurity**
Google · May 2024 · Credential ID: QNNS7VZR384S
Verify Certificate

LANGUAGES

Bengali	Native / Bilingual
English	Professional Working Proficiency – IELTS Band 7.0 / C1 [Certificate]
Hindi	Conversational
German	Elementary (A1 – Currently Learning)

HOBBIES & INTERESTS

Vocal performance and appreciation of diverse musical genres, encompassing both classical and contemporary compositions. Avid reader of literary fiction, non-fiction, and academic literature spanning technology, science, and social sciences. Enthusiast of international television series and cinematic productions, with a particular interest in narrative storytelling, character development, and cross-cultural perspectives.

PERSONAL INFORMATION

Nationality:	Bangladeshi
Date of Birth:	29 December 2002